

AYMANE AIT DADS

Test-Time Adaptation Research Intern | PyTorch · TTA · Robust AI

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Hiring Team - Internship | Test-Time Adaptation for Robust AI

TNO · The Hague, Netherlands

Dear TNO Hiring Team,

Your internship brief asks for applied research on **Test-Time Adaptation** to make models robust when real-world conditions shift at inference time - exactly the problem I attacked in my **Robust AI-Generated Image Detection** project at EURECOM, where I ranked **#1 on the private leaderboard** by designing a **10-view test-time augmentation** strategy that preserved out-of-distribution generalization across 25+ unseen generator types without any additional training.

Two results speak directly to TNO's research directions: first, tackling **adaptation on as little data as possible** - my ablation experiments proved that **1-epoch fine-tuning** of a **428M-parameter CLIP ViT-L/14** on **250K samples** generalized better than extended re-training, showing that less adaptation can outperform more. Second, on **optimal use of compute resources** - I implemented **FP16 mixed precision with GradScaler** to run the full pipeline within a constrained T4 GPU budget, exactly the kind of efficiency constraint relevant to cheap hardware and embedded deployment scenarios TNO describes.

TNO's focus on deploying adaptive AI in military-relevant scenarios - autonomous vehicles, drones, degraded sensor conditions - is a research context where stability and compute constraints are not academic concerns but operational requirements. I would welcome the opportunity to discuss how my work maps to your current TTA research directions.

Sincerely,

AYMANE AIT DADS